Ratio & Proportion

Year 4 · **Lesson Worksheet**

Worked Examples

Easy: A recipe needs 2 eggs for every 1 cake. How many eggs are needed for 3 cakes?

Answer: 2 eggs × 3 = 6 eggs.

Standard: Share 12 sweets in the ratio 1:2.

Answer: 1+2=3 parts. 12÷3=4 per part. So the shares are 4 and 8.

Advanced: A recipe for 4 people needs 200g of flour. How much flour is needed for 6 people?

Answer: 200g ÷ 4 = 50g per person; 50g × 6 = 300g.

Easy / Standard (15 questions)

1. A recipe for 1 person needs 7 biscuits. How many biscuits for 2 people?
2. Scale up the ratio 3:4 by multiplying both parts by 2. What is the new ratio? A) 5:6 B) 6:7 C) 6:8 D) 3:8
3. A pattern uses 4 blue beads for every 1 red bead. If there are 3 red beads, how many blue beads are there?
4. Share 8 sweets equally between 2 people. Select ALL statements that are TRUE: A) Each person gets 4 sweets B) Each person gets 3 sweets C) This is sharing in the ratio 1:1 D) This is sharing in the ratio 2:1 E) Together they get 8 sweets
5. A recipe needs 2 eggs for every 1 cake. How many eggs are needed for 4 cakes?
6. A recipe for 1 person needs 8 biscuits. How many biscuits for 4 people?
7. Scale up the ratio 2:4 by multiplying both parts by 4. What is the new ratio? A) 6:8 B) 8:16 C) 8:12 D) 2:16
8. A pattern uses 3 blue beads for every 1 red bead. If there are 4 red beads, how many blue beads are there?
9. Share 6 sweets equally between 2 people. Select ALL statements that are TRUE: A) Each person gets 3 sweets B) Each person gets 2 sweets C) This is sharing in the ratio 1:1 D) This is sharing in the ratio 1:2 E) Together they get 6 sweets
10. A recipe needs 3 eggs for every 1 cake. How many eggs are needed for 3 cakes?
11. A recipe for 1 person needs 2 biscuits. How many biscuits for 4 people?
12. Scale up the ratio 4:3 by multiplying both parts by 4. What is the new ratio? A) 8:6 B) 16:3 C) 16:12 D) 12:16
13. A pattern uses 3 blue beads for every 1 red bead. If there are 5 red beads, how many blue beads are there?
14. Share 4 sweets equally between 2 people. How many does each get?
15. A recipe needs 3 eggs for every 1 cake. How many eggs are needed for 4 cakes?

Standard (15 questions)

1. In a fruit bowl, apples and oranges are in the ratio 1:1. There are 4 pieces of fruit in total. How many apples are there?

© PassFrame Tuition — for registered student use

2. A recipe for 1 person uses 2 grams of butter. How much butter is needed for 5 people? A) 5g B) 7g C) 10g D) 12g

3. A recipe for 3 people uses 18g of sugar. How much sugar is needed for 6 people?

4. Share 20 counters in the ratio 2:2. Select ALL statements that are TRUE: A) Each share is 10 B) The shares are equal C) One share is 15 and the other is 5 D) This is the same as sharing equally between 2 people E) One share is 12 and the other is 8

5. A model car is built at a scale where 1 cm represents 17 cm in real life. If the model is 4 cm long, how long is the real car?

6. In a fruit bowl, apples and oranges are in the ratio 1:2. There are 15 pieces of fruit in total. How many apples are there?

7. A recipe for 1 person uses 5 grams of butter. How much butter is needed for 8 people? A) 13g B) 35g C) 40g D) 45g

8. A recipe for 3 people uses 9g of sugar. How much sugar is needed for 9 people?

9. Share 24 counters in the ratio 2:2.

10. A model car is built at a scale where 1 cm represents 18 cm in real life. If the model is 4 cm long, how long is the real car?

11. In a fruit bowl, apples and oranges are in the ratio 2:1. There are 12 pieces of fruit in total. How many apples are there?

12. A recipe for 1 person uses 5 grams of butter. How much butter is needed for 3 people? A) 8g B) 15g C) 20g D) 25g

13. Share 9 counters in the ratio 1:2. Select ALL statements that are TRUE: A) The shares are 3 and 6 B) The shares are 4 and 5 C) There are 3 parts in total D) There are 2 parts in total E) The larger share is 6

14. A model car is built at a scale where 1 cm represents 20 cm in real life. If the model is 2 cm long, how long is the real car?

15. A recipe for 2 people uses 5 grams of butter. How much butter is needed for 6 people?

Standard / Hard (15 questions)

1. A recipe for 2 people needs 8g of pasta. How much pasta is needed for 6 people? A) 12g B) 16g C) 24g D) 32g

2. 3 pencils cost £9. Using the same price per pencil, how much would 5 pencils cost?

3. A tap fills a container at a steady rate. In 2 minutes it fills 6 litres. Using the same rate, how many litres will it have filled after 5 minutes?

4. Share 40 pens in the ratio 3:5. How many are in the larger share?

5. A drink mixes juice and water in the ratio 1:3. If there are 20ml altogether, how much is juice and how much is water?

6. A recipe for 2 people needs 10g of pasta. How much pasta is needed for 5 people? A) 15g B) 20g C) 25g D) 30g

7. 3 pencils cost £15. Using the same price per pencil, how much would 6 pencils cost?

8. A tap fills a container at a steady rate. In 3 minutes it fills 9 litres. Select ALL statements that are TRUE: A) The rate is 3 litres per minute B) After 5 minutes it will have filled 15 litres C) The rate is 4 litres per minute D) After 1 minute it fills 3 litres E) After 5 minutes it will have filled 20 litres

9. Share 30 pens in the ratio 2:3. How many are in the larger share?

10. A drink mixes juice and water in the ratio 1:2. If there are 9ml altogether, how much is juice and how much is water?

© PassFrame Tuition — for registered student use

11. A recipe for 2 people needs 8g of pasta. How much pasta is needed for 7 people? A) 24g B) 28g C) 32g D) 36g

12. 3 pencils cost £9. Using the same price per pencil, how much would 8 pencils cost?

13. A tap fills a container at a steady rate. In 2 minutes it fills 8 litres. Using the same rate, how many litres will it have filled after 6 minutes?

14. Share 20 pens in the ratio 1:4. How many are in the larger share?

15. A drink mixes juice and water in the ratio 2:3. If there are 20ml altogether, select ALL statements that are TRUE: A) There are 5 parts in total B) There is 8ml of juice C) There is 12ml of water D) There is 10ml of juice E) Each part is 4ml

© PassFrame Tuition — for registered student use